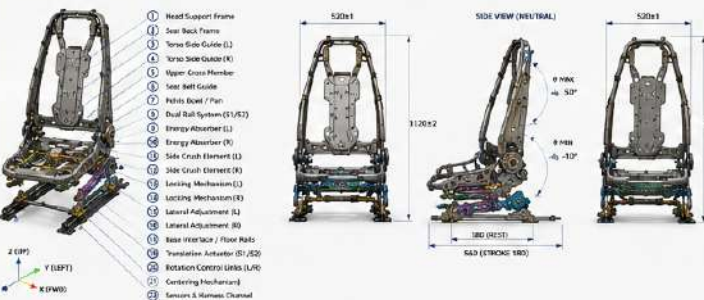


T-OCS V6.0 – TIME-PROGRAMMABLE OCCUPANT CONTROL SEAT

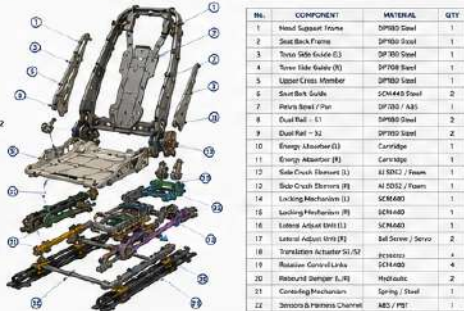
ARCHITECTURAL CAD MODEL – REFERENCES, JOINTS, LOAD PATHS & STATES

DUAL LOAD PATHS • DECOUPLED ROTATION CONTROL • PROGRESSIVE ENERGY ABSORPTION • MULTI-STATE LOCK • REBOUND CONTROL

1. SYSTEM ARCHITECTURAL OVERVIEW



2. MAJOR ASSEMBLY – EXPLODED VIEW (REFERENCE)



3. STATE MACHINE – 6-STATE CONTROL LOGIC

STATE ID	EVENT NAME	DESCRIPTION	KEY ACTIONS
S0 NORMAL (DRIVING)	All systems free	Seat adjustable	All locks open / absorbers inactive
S1 PREPARED (ARMED)	System armed	Trigger initiated	Pre-tension dampers / Fireable trigger logic
S2 HIGH G (CRASHING)	Pelvis secured	Guides engaged	Pelvis bowl locks / Side guides close
S3 ABRUPT (BRAKE)	Seat translates	Energy absorbed	Absorbers active
S4 REBOUND CONTROL	Return controlled	Prevent rebound	Rebound dampers
S5 END STATE (SECURE)	System locked	Safety state	Final locks engage

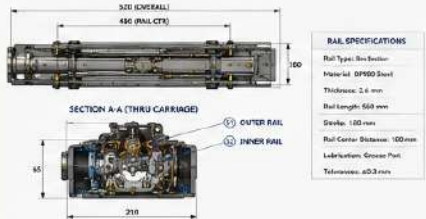
STATE TRANSITION CONDITIONS

- Initial trigger: accel threshold (L_z or A_z)
- On Seat position / seat back angle change
- On L_z / A_z / vehicle-pulse detection
- On Fire / E-trigger (if equipped)

SAFETY DEFAULT

If trigger or power fails → system reverts to SAFE (S0/S1)
Primary restraint path always available

4. DUAL RAIL SYSTEM (S1 / S2) – PLAN & SECTION



5. LOAD PATH DIAGRAM – DUAL PATH ARCHITECTURE



7. ENERGY ABSORBER CARTRIDGE – V6.0



PERFORMANCE (PER SIDE)	
Peak Force:	18 – 22 kN
Stroke:	8 – 180 mm
Energy Absorption:	2.0 – 2.8 kJ
Stroke:	2.0 – 2.8 kJ
Avg. Force:	11 – 16 kN
Operating Temp:	40°C to -80°C
Life Cycle:	15 Years / 200K Cycles

8. TRAJECTORY ENVELOPE & KINEMATIC CONTROL



9. REAR SEAT CONFIGURATIONS – V6.0



10. VEHICLE INTERFACE & ANCHORS



11. TECHNICAL SPECIFICATIONS (NOMINAL)

Total Weight (per seat)	29.8 kg ± 1.0 kg
Total Stroke (mm)	160 mm
Seat Back Rotation (max)	± 30°
Seat Pan Pitch Range	10° to +25°
Peak Belt Load (per seat)	1.77 kN
System Lifetime	15 Years / 200K Cycles
Dynamic Load Capacity	> 110 kg
Life Cycle	15 Years / 200K Cycles

12. TOLERANCE TABLE (UNLESS SPECIFIED)

Linear Dimension (30–100)	±0.3 mm
Linear Dimension (100–200)	±0.5 mm
Angular Dimension	±0.5°
Hole Dimension (H7)	
Shaft Dimension (h7)	
Perforation	0.02 mm / 0.01 mm
Perforation	0.05 mm / 0.03 mm
Concentricity	0.05 mm

13. VALIDATION & TEST MATRIX

TEST LEVEL	METHOD	OBJECTIVE	PASS CRITERIA
Component	Static Test	Assembly / Load	Force / Stroke within bounds
Subsystem	Static Test	Assembly / Load	Within design envelope
CAE / FEM	Simulation	Corrosion	< 0.1% strain in load
Full Vehicle	Crash Test	Safety Performance	Meets targets / Regs.
Durability	Cycle Test	Life / Wear	200K cycles OK

14. MATERIALS & FINISH

COMPONENT GROUP	MATERIAL	FINISH / TREATMENT
Frame / Rails	DP100 / DP100	E-Coat + Powder
Cross Members	DP100	E-Coat + Powder
Pan / Shafts	SENA40	Q&T + Black Oxide
Energy Absorber Housing	Aluminum / Steel	Anodized / F.S. Coat
Springs	SLP10	Shot Peened + Oil
Buildings	Stainless Steel	Citriprepassant

15. KEY ADVANTAGES – V6.0

- ✓ Decoupled translation & rotation control
- ✓ Redundant left/right load paths
- ✓ Progressive energy absorption (multi-stage)
- ✓ Six-state control logic with fail-safe
- ✓ Explicit rebound control
- ✓ Rebound ramping & secondary impacts
- ✓ Serviceable / modular cartridges
- ✓ FEA, S&B & Full-vehicle validated roadmap

15. REVISION HISTORY

REV.	DATE	DESCRIPTION	BY
A	2023-09-10	Initial Concept Release	ENG
B	2023-07-10	Added Detailed Mechanisms	ENG
C	2023-06-10	Added Crash Analysis & Claims	ENG
D	2023-01-01	V6.0 – Optimized & Finalized	ENG

T-OCS V6.0



TIME-PROGRAMMABLE CONTROL



DUAL LOAD PATHS



DECOUPLED ROTATION CONTROL



PROGRESSIVE ABSORPTION



REBOUND & FAIL-SAFE



ACTIVE TRAJECTORY MANAGEMENT



MODULAR & SERVICEABLE

STATUS: RESEARCH PROTOTYPE
NOT CERTIFIED FOR PRODUCTION
All dimensions are nominal. Validate with CTD & tools.
This document is for project preparation only.

T-OCS V6.0 – TIME-PROGRAMMABLE OCCUPANT CONTROL SEAT
NEXT-GENERATION MULTI-STATE DYNAMIC SEATING SYSTEM FOR REAR IMPACT PROTECTION



DESIGN TARGET (Example - Rear Impact 25.6 g)

Total Stroke (Seat Travel)	180 mm
Peak Deceleration Managed	16 - 22 g
Energy Absorption	2.0 - 2.8 kJ
Seat Back Rotation (Max)	5.25°
Operating Temp.	-40°C to +80°C
Life Cycle	15 Years / 200K Cycles

- KEY FEATURES – V6.0 ADVANCEMENTS
- Occupied seat translation & swivel/rotation control
 - Dual-rail redundant load paths (fail-ongoing)
 - Multi-stage energy absorber cartridge (L→M→H)
 - 6 State lock & control logic with rebound management
 - Adaptive torso guides with load distribution surfaces
 - Proven torsion system patent
 - Serviceable modular architecture
 - Crash-ready design validated by FE, Sled & Full-Vehicle tests

EFFECTIVE FORCE CONCEPT (INERTIAL FRAME)

$$F_{eff}(t) = m(g - a_{seat}(t))$$

NOT a gravity generator – an inertial control strategy

1. OVERVIEW – V6.0 SYSTEM ARCHITECTURE



2. EXPLODED CAD – MAJOR ASSEMBLY



NO.	COMPONENT	MATERIAL	QTY
1	Head Support Frame	DP600 Steel	1
2	Headrest & Wing	DP780 Steel	1
3	Torso Side Guide (L)	DP580 Steel	1
4	Torso Side Guide (R)	DP580 Steel	1
5	Torso Cross Member	DP580 Steel	1
6	Back Panel (Compliance)	DP580 Steel	1
7	Back Hinge (L)	SCM440 Steel	1
8	Back Hinge (R)	SCM440 Steel	1
9	Pelvis Seat (Anti-Submerging)	DP580 Steel	1
10	Pelvis Adjust Actuator (L)	SCM440 Is.	1
11	Pelvis Adjust Actuator (R)	SCM440 Is.	1
12	Seat Pan (Mayday)	DP580 Steel	1
13	Pan Front Hinge (L)	SCM440 Steel	1
14	Pan Front Hinge (R)	SCM440 Steel	1
15	Dual Rail – S1 (Main)	DP580 Steel	2
16	Dual Rail – S2 (Redundant)	DP580 Steel	2
17	Translation Actuator (S1/S2)	SCM440 Is.	2
18	Energy Absorber – Cushion	0941-T9 Al.	2
19	Energy Absorber – Cush Element	5051 Al.	2
20	Base Frame (High Strength)	DP580 Steel	1
21	Lateral Adjust Unit (N)	SCM440 Steel	1
22	Locking Mechanism (Ratchet)	SCM440 Steel	1
23	Status Indicator (Green/Red)	PCB/D	1
24	Washer & Control (Bracket)	AN/PCL	1

3. SYSTEM STATES – 6-STATE CONTROL LOGIC

STATE	NAME	STATUS	KEY ACTION
50 Normal Driving	UNLOCKED	UNLOCKED	Free adjustment / Comfort
S1 Prepare (Armed)	Armed	ARMED	Trigger enabled / Pre-occupy
S2 Pelvis Lock	Paths	CAPTURE	Pelvis secured / Guides engage
Roll-Down (Brug)	(Roll-Down)	ROLL-DOWN	Seat translates / Absorbers active
S4 Rebound/Control	Control	REBOUND	Dampers control return motion
End-Stop / Secure	LOCKED	LOCKED	Final locks engaged / Safe state

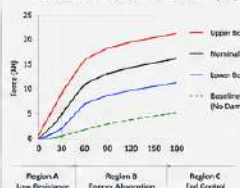
STATE TRANSITION TRIGGERS

- Inertial trigger (Acceleration threshold)
- OR Sensor fusion (Seat accel + vehicle accel)
- OR Seat position + time window logic
- OR Pyro / B-Trigger (if equipped)

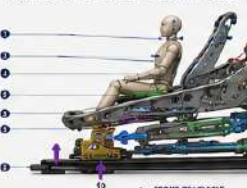
4. ENERGY ABSORBER CARTRIDGE – V6.0



ABSORBER FORCE-STROKE CURVE (mm)



5. LOAD PATH – FROM OCCUPANT TO VEHICLE



6. TRAJECTORY MANAGEMENT



7. REAR SEAT CONFIGURATIONS (V6.0)



8. CRASH LOAD PATH TIMELINE (0-400 ms)



9. MATERIAL STRATEGY

COMPONENT	MATERIAL	PROPERTY
Seat Back Frame	DP580 / Hot-Dipped	High Strength
Seat Pan	DP780 / AISI	High Strength
Rails	22MnB5 / QBP Steel	High Strength
Cross Members	QBP99	High Strength
Energy Absorber Housing	High Strength Steel	Energy Absorption
Crash Element Core	Aluminum / Polymer	Energy Absorption
Fasteners	SCM440 / 12.9 Grade	High Strength
Bushings	Stainless Steel	Wear Resistance

10. VALIDATION PLAN (V6.0)

PHASE 1	PHASE 2	PHASE 3	PHASE 4
Simulation	Component	Sled Test	Full Vehicle
CAE / MBD	Absorber / Lock	THOR-50M / BURD-6	Rear Impact
S DYNA / Abaqus	Bench Tests	High-Speed Video	FMVSS 207 / 301
Param. Studies	Durability	Load Cells	Validation

11. PERFORMANCE TARGETS (GUIDANCE)

- ✓ Seat Back Rotation ≤ 5.25°
- ✓ Seat Translation ≤ 100 mm
- ✓ Peak Seat Load ≤ 15 kN
- ✓ Pelvis Acceleration (Peak) ≤ 12 g
- ✓ Energy Absorption 2.0 - 2.8 kJ
- ✓ Rebound Travel ≤ 10 mm

12. KEY BENEFITS (V6.0)

- ✓ Reduces rear impact injury risk
- ✓ Controls occupant trajectory
- ✓ Limits seat back collapse & rotation
- ✓ Distributes loads through dual paths
- ✓ Manages rebound & secondary impact
- ✓ Modular, serviceable & scalable

13. APPLICABILITY



• This document is a research prototype disclosure.
• All dimensions are nominal unless otherwise stated.
• Not for production without validation.
• Patent claims must be reviewed by a patent attorney.

REVISION HISTORY

REV	DATE	DESCRIPTION	BY
A	2025-09-10	Initial Concept Review	ENG
B	2025-07-10	Added Detailed Mechanisms	ENG
C	2025-06-10	Added Crash Analysis & Claims	ENG
D	2025-09-02	V6.0 - Optimized & Finalized	ENG

T-OCS V5.0 – ENERGY ABSORBER MODULE

EXPLODED CAD – DETAILED DESIGN & ENGINEERING PACKAGE

Progressive Energy Absorber with Damping Cartridge & Mounting Interfaces

Units: mm | Scale: NTS

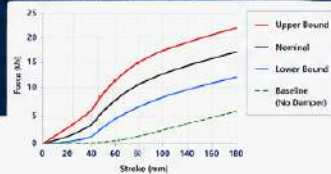
1. EXPLODED VIEW – ENERGY ABSORBER MODULE



DESIGN TARGET (Example – 25.6 g Pulse)

Total Stroke	180 mm
Peak Force	18 – 22 kN
Energy Absorption	2.0 – 2.8 kJ
Operating Temp.	-40°C to +80°C
Oil Type	ISO VG 15 – 22
Gas Precharge	3 – 5 bar (N ₂)

2. FORCE-STROKE CHARACTERISTIC (TYP.)



3. MOUNTING INTERFACES



4. LOAD PATH – ENERGY FLOW



- Impact / Seat Back Load
- Carriage Transfer (Cross Members)
- Absorber Compression (Oil & Spring)
- Force Transmission to Rails
- Rail to Floor Load
- Vehicle Structure / Floor
- Reaction to Vehicle Floor

5. SPECIFICATIONS

Max Stroke (Total)	180 mm
Peak Force (Nominal)	20 kN
Energy Absorption	2.0 – 2.8 kJ
Average Force	11 – 16 kN
Operating Temperature	-40°C to +80°C
Oil Viscosity	15 – 25 cSt @ 40°C
Life Cycle	15 Years / 200K Cycles

6. NOTES

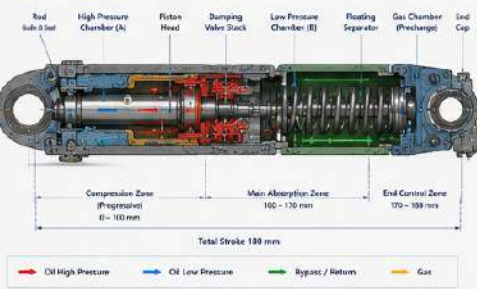
- All dimensions are in mm unless stated.
- Symmetric modules: Left & Right.
- Cartridge is serviceable and replaceable.
- Use only recommended oil & seal kit.
- Do not disassemble cartridge without clean room procedure.
- Validate with FEA + Sted Tests.
- For production, verify GD&T per full drawing set.

T-OCS V5.0 – ENERGY ABSORBER MODULE

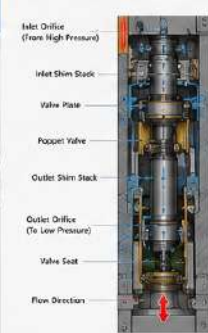
DEEP CROSS SECTION & INTERNAL DYNAMIC FLOW

Units: mm | Scale: NTS

1. LONGITUDINAL CROSS SECTION (CENTER CUT)



2. VALVE STACK – DETAILED SECTION (BLOW-OFF TYPE)

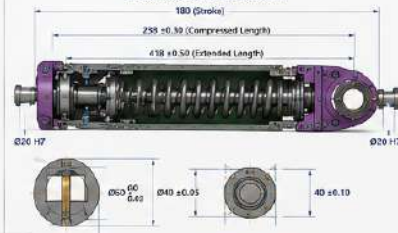


3. SHIM STACK CONFIGURATION (TYP.)



Shim Stack	ID (mm)	OD (mm)	T (mm)	QTY	Material
Inlet Stack	10.2	24.0	0.10 / 0.20	6	17-4PH SS
Outlet Stack	10.2	24.0	0.10 / 0.20	6	17-4PH SS

4. DIMENSIONS & TOLERANCES



FEATURE	TOLERANCE	DATUM	NOTE
Clevis Pin Holes (Ø20 H7)	Ø0.021	A-B	Both Ends
Clevis Face	0.10	A	Perpendicularity
Mounting Face Flatness	0.10	A	Both Ends
Rail Mounting Holes Position	Ø0.20	A-B	2x
Piston Rod Runout	0.05	A	At Full Stroke
Overall Length	438.50	—	Extended

5. TOLERANCES (UNLESS SPECIFIED)

Linear Dimensions	±0.10 mm
Angles	±0.5°
Hole H7	As per ISO 286
Shaft h6	As per ISO 286
Parallelism	0.10 mm
Perpendicularity	0.10 mm
Concentricity	0.05 mm
Surface Finish	Ra ≤ 1.6 µm

6. SERVICEABLE CARTRIDGE

Cartridge can be removed as a unit for replacement / overhaul.



T-OCS V5.0 – ENERGY ABSORBER MODULE

EXPLODED CAD – DETAILED DESIGN & ENGINEERING PACKAGE

Progressive Energy Absorber with Damping Cartridge & Mounting Interfaces

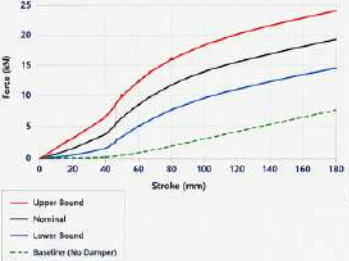
1. OVERVIEW – EXPLODED ASSEMBLY



No.	Component	Material / Spec	Finish	Qty	Function	No.	Component	Material / Spec	Finish	Qty	Function
1	Rod End (Seat Side)	DP980 Steel	E-Coat	1	Connects to seat carriage	12	Spacer Ring	303S-Ti	Acetone	1	Seals piston rod
2	Spherical Bearing	SAE4140 Steel	PTFE Liner	1	Allows angular misalignment	13	Main Coil Spring	SUP10	Shot Peened	1	Provides progressive force
3	Shim Ring	SAE4140 – 60	Spring Peened	1	Retains bearing	14	Spring Seat (Inner)	303S-Ti	Anodized	1	Seats spring
4	Rod Guide / Bushing	Stainless Bronze + PTFE	–	1	Guides piston rod	15	Cartridge Tube	DP980 Steel	E-Coat	1	Pressure housing
5	Piston Rod	A204M4 / Hard Chrome	20–25 µm	1	Transmits force	16	Flange Separator	POM (Acetal)	–	1	Separates oil & gas
6	Front Seal (Rod)	PU / NBR	–	1	Prevents leakage	17	Oil Seal (Cartridge)	NBR	–	1	Seals cartridge
7	Piston Head	Aluminum 6061-T6	Hard Anodized	1	Carries valve stack	18	End Cap (Rod Side)	DP980 Steel	E-Coat	1	Closes housing
8	Wear Ring	POM (Acetal)	–	1	Reduces friction	19	O-Ring (End Cap)	NBR	–	1	Static seal
9	Seal Ring (Piston)	PU / NBR	–	1	High pressure seal	20	Mounting Clevis (Ball)	SCM440 / 10.9	Powder Coat	1	Connects to rail
10	Valve Stack (See Detail B)	–	–	1	Controls flow & damping	21	Mounting Bolt	M12 x 1.25	–	2	Fastens to clevis
11	Backstop Ring	PTFE	–	1	Supports seal under pressure						

* All metals free of burrs & sharp edges. All seals lubricated with MOLYKOTED P-74.
** Serviceable cartridge design – can be replaced as a module.

4. FORCE–STROKE CHARACTERISTIC (TYP.)



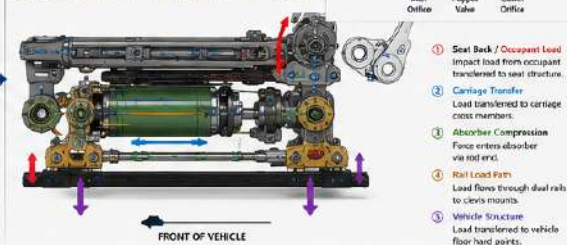
ZONES

- 21 0 – 20 mm Initial Take-Up
Low force for smooth ride-down
- 22 20 – 100 mm Progressive Absorption
Force increases with controlled slope
- 23 100 – 160 mm High Energy Absorption
Maximizes peak force
- 24 160 – 180 mm End-Control Zone
Limits maximum stroke & force

5. DEFORMATION / OPERATION STAGES



6. LOAD PATH – FROM SEAT TO VEHICLE STRUCTURE



7. IMPORTANT DIMENSIONS (mm)



8. TOLERANCES (UNLESS SPECIFIED)

DESCRIPTION	TOLERANCE
Linear Dimensions (0 – 100 mm)	±0.10 mm
Linear Dimensions (100 – 300 mm)	±0.15 mm
Linear Dimensions (300 – 1000 mm)	±0.30 mm
Hole (H7)	As per ISO 286
Shaft (h6)	As per ISO 286
Parallelism	0.10 mm
Perpendicularity	0.10 mm
Concentricity	0.10 mm
Surface Finish	Ra ≤ 1.6 µm

9. CARTRIDGE SERVICE DATA



10. MATERIAL & FINISH

COMPONENT	MATERIAL	FINISH / TREATMENT
Rod End, End Cap	DP980 Steel	E-Coat + Powder
Piston Rod	42CrMo4	Hard Chrome 20–25 µm
Piston Head	Aluminum 6061-T6	Hard Anodized
Cartridge Tube	DP980 Steel	E-Coat
Clevis (Rail)	SCM440	Powder Coat
Valve Stack Parts	17-4PH / H900	Passivated
Springs	SUP10	Shot Peened + Oil

11. NOTES

- 1. All dimensions in mm unless stated.
- 2. Use only recommended oil & seal kit.
- 3. Do not disassemble cartridge.
- 4. Validate performance with FEA & sled tests.
- 5. Design for multi-hit; cool down time > 30 s.
- 6. This module is symmetric (Left & Right).
- 7. For production, verify GCRT per full drawing set.

KEY SPECIFICATIONS (Example – 25.6 g Pulso)

Total Stroke	180 mm
Peak Force	18 – 22 kN
Energy Absorption	2.0 – 2.8 kJ
Operating Temp.	-40°C to +80°C
Oil Type	ISO VG 15 – 22
Gas Precharge	3 – 5 bar (N ₂)

KEY FEATURES

- ✓ Progressive force characteristic
- ✓ Low initial force – smooth ride-down
- ✓ High energy absorption – compact
- ✓ Integrated damping cartridge
- ✓ Serviceable & replaceable
- ✓ Symmetric dual-rail compatible
- ✓ High strength – multi-hit robust

MOUNTING INTERFACES



MOUNTING BOLTS

Bolt Size	M12 x 1.25
Grade	ISO 10.9
Tightening Torque	80 ±5 N·m
Thread Engagement	≥ 1.5 x Dr
Washer	DIN 125 – M12

FLOW SCHEMATIC (DAMPING)



T-OCS V5.0 – ENERGY ABSORBER MODULE

DETAILED MECHANICAL DEVELOPMENT

Progressive Energy Absorption • Controlled Ride-Down • Rebound Management

DESIGN TARGET (Example – 25.6 g Pulse)

Stroke (Total)	180 mm
Peak Force	18 – 22 kN
Energy Absorption	2.0 – 2.8 kJ
Operating Temp.	-40°C to +80°C

KEY FEATURES

- ✓ Progressive force characteristic
- ✓ Serviceable cartridge (replaceable)
- ✓ Symmetric dual absorbers (L/R)
- ✓ Integrated rebound damping
- ✓ High energy capacity, compact
- ✓ Designed for multi-hit robustness

LEGEND – LOAD PATH

- ➔ Impact / Seat Back Load
- ➔ Shear Load
- ➔ Compression Load
- ➔ Reaction to Vehicle Floor

SYSTEM CONTEXT

Energy absorber modules (Left & Right) connect the seat carriage to the dual rails. Load is transferred through the modules to the vehicle floor hard points.

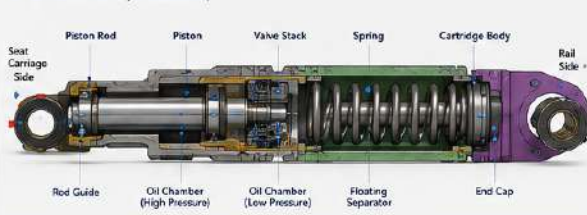


1. ENERGY ABSORBER MODULE – EXPLODED VIEW



No.	Component	Material / Spec.	Function	No.	Component	Material / Spec.	Function
1	End Clevis (Seat Side)	DP980 Steel	Connects to Seat Carriage	7	Main Coil Spring	SUP10	Supports ride-down, stores energy
2	Rod Guide	S45C / Hard Chrome	Guides rod, reduces side load	8	Cartridge Body	Aluminum 6061-T6	Contains oil & components
3	Seal Assembly (Rod)	PU / NBR	Prevents oil leakage	9	End Clevis (Rail Side)	DP980 Steel	Connects to Dual Rail
4	Piston Rod	420-Mk4 / Hard Chrome	Transmits force	10	Mounting Bolt	SCM440 / 10.9	Fastening to clevis
5	Piston	S45C	Converts hydraulic force	11	Bush / Bearing	Self-lubricating	Allows rotation, reduces wear
6	Valve Stack (Shim)	Spring Steel	Progressive flow control				

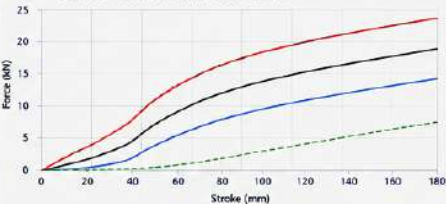
2. SECTION VIEW (ASSEMBLED)



4. ENERGY ABSORPTION – DEFORMATION STAGES



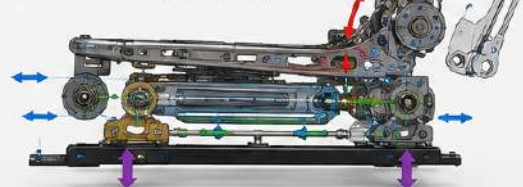
3. FORCE-STROKE CHARACTERISTIC



CHARACTERISTIC ZONES

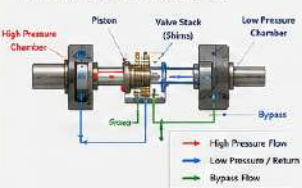
- Z1 0 – 20 mm Initial Take-Up Low force for smooth ride-down
- Z2 20 – 100 mm Progressive Absorption Force increases with controlled slope
- Z3 100 – 160 mm High Energy Absorption Maximum peak load
- Z4 160 – 180 mm End-Control Zone Limits maximum stroke & force

5. LOAD PATH – FROM SEAT TO VEHICLE STRUCTURE



- ➔ **Seat Back Load**
Impact load from occupant transfers to seat structure.
- ➔ **Carriage Transfer**
Load transferred to seat carriage cross members.
- ➔ **Absorber Compression**
Force enters absorber through end clevis (seat side).
- ➔ **Force Transmission**
Load transferred through absorber to rail side clevis.
- ➔ **Rail to Floor**
Load flows through dual rails to vehicle floor hard points.

6. HYDRAULIC CIRCUIT (INTERNAL)



7. REBOUND (RETURN) CONTROL



During rebound, oil flows through check valve and rebound orifice. Flow is restricted to control return speed. Prevents seat from rebounding and throwing occupant forward.

8. PERFORMANCE SUMMARY

Parameter	Value (Example)	Unit	Note
Total Stroke	180	mm	Per Side
Peak Force	18 – 22	kN	At 25.6 g Pulse
Energy Absorption	2.0 – 2.8	kJ	Per Side
Average Force	11 – 16	kN	Over Stroke
Operating Temp.	-40 to +80	°C	Validated
Oil Viscosity	15 – 25	cSt	@ 40°C
Life Cycle	15 Years / 200K	Cycles	Design Target

9. MOUNTING & INTERFACE DETAILS



- #### 10. NOTES
- All dimensions are in mm unless stated.
 - Symmetric modules: Left & Right.
 - Serviceable cartridge design.
 - Validate with FEA + Sled Test.
 - Do not rely on single test result.
 - Customize for each vehicle platform.
- #### 11. VALIDATION PLAN
- Component Test (Force-Stroke)
 - Thermal Test (-40°C to +80°C)
 - Durability Test (200K cycles)
 - Sled Test (25.6 g Pulse)
 - Full Vehicle Validation
 - FMVSS / ECE Compliance

T-OCS V5.0 – EOK/UNLOCK MECHANISM

DEEP MECHANICAL DEVELOPMENT

Multi-State Inertial Locking System for Time-Programmable Ride-Down Control

1. LOCKING MODULE – EXPLODED VIEW



No.	Component	Material / Spec	Qty	Notes
1	Housing	DP980 / 2.0t	1	E-coated
2	Drive Cam	S45C / HRC 45-50	1	Precision machined
3	Pawl	SCM415 Carburized	1	Heat treated
4	Return Spring	SAE 1074	1	Shot peened
5	Lock Pin	SCM440 / Ø16	1	Press fit
6	Status Indicator	Polycarbonate	1	Green/Red
7	Release Lever	DP980 / 2.5t	1	Pivoted
8	Inertial Mass	Steel / 120 g	1	Sliding
9	Tension Spring	SUS301	1	For lever return
10	Damping Pad	Polyurethane	1	Vibration damping

FUNCTION OVERVIEW

The Locking System secures the seat structure during normal driving. In a rear impact event, inertial trigger mass moves, rotating the cam and releasing the pawl. The mechanism transitions through PROHIBIT → RIDE-DOWN → END-STOP states to control motion and then re-locks after the event.

KEY COMPONENTS

- ① Housing (DP980 Steel)

② Drive Cam (S45C Steel)

③ Pawl (Carburized Steel)

④ Return Spring (SAE 1074)

⑤ Lock Pin (SCM440)
- ⑥ Status Indicator (DP980)

⑦ Release Lever (DP980)

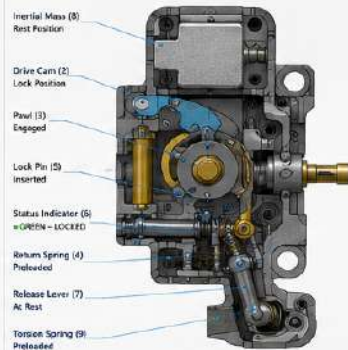
⑧ Inertial Trigger Mass

⑨ Tension Spring (SUS301)

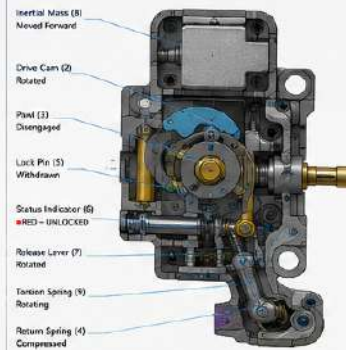
⑩ Damping Pad (PU)



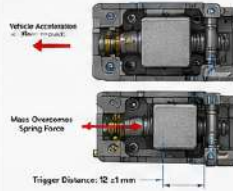
2. CROSS SECTION (LOCKED STATE)



3. CROSS SECTION (UNLOCKED – RIDE-DOWN STATE)



4. INERTIAL TRIGGER MECHANISM

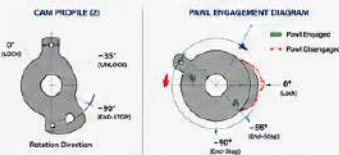


TRIGGER SPECIFICATION	
Trigger Acceleration	≥ 2.5 g
Engage Time (Unlock)	< 20 ms
Reset Acceleration	> -1.0 g
Reset Time (Relock)	< 300 ms
Life Cycle	15 Years / 200k Cycles
Operating Temp.	-40°C to +80°C

5. LOCK/UNLOCK SEQUENCE – MECHANISM STATES



6. CAM PROFILE & PAWL ENGAGEMENT



7. TYPE OF LOCKING – MULTI-REDUNDANT



8. MOUNTING & INTERFACE



9. LOAD PATH THROUGH LOCKING SYSTEM



10. MATERIAL & FINISH

COMPONENT	MATERIAL	FINISH / TREATMENT
Housing	DP980	E-Coat + Powder
Drive Cam	S45C	Q&T + Black Oxide
Pawl	SCM415	Carburized + QPC
Lock Pin	SCM440	Hardened + Zn-Ni
Spring	SAE1074 / SUS301	Shot Peened
Lever	DP980	E-Coat + Powder
Inertial Mass	Steel	Zn-Ni Plated

11. TOLERANCES (UNLESS SPECIFIED)

Linear Dimension	0.1 – 0.3 mm
Hole Diameter	H7
Shaft Diameter	h6
Angle	± 0.5°
Parallelism	0.1 mm
Perpendicularity	0.1 mm
Concentricity	0.1 mm

12. NOTES

- All fasteners: Class 10.9 or equivalent.
- Lubricate cam, pawl, and pin with MOLYKOTE P-74.
- Test per life cycle and environmental requirements.
- Full-safety design: defaults to LOCKED when no trigger.
- Serviceable modular unit.

13. KEY BENEFITS

- ✓ Fast inertial unlock (< 20 ms)
- ✓ Positive mechanical lock in all states
- ✓ Redundant safety (cam + pin + brackets)
- ✓ High-strength materials for crash loads up to 25.6 g
- ✓ Automatic relock after event
- ✓ Low NVH with damping pad

14. OPERATION PRINCIPLE (OVERVIEW ANIMATION STRIP)



15. TEST & VALIDATION PLAN

- Inertial trigger test (2.5 g to 10 g)
- Durability test (200k cycles)
- Temperature test (-40°C to +80°C)
- Shed test with 25.6 g pulse
- Corrosion test (96 h salt spray)
- Drop test (1.5 m)
- Vibration test (5-2000 Hz)

T-OCS V5.0 – ENERGY ABSORPTION & LOCK/UNLOCK MECHANISM

EXPLODED CAD – DETAILED DESIGN PACKAGE

Time-Programmable Effective Gravity Seat – Rear Impact Protection

1. SYSTEM OVERVIEW – EXPLODED ASSEMBLY

1 Head Support Frame
2 Seat Back Frame
3 Upper Cross Member
4 Torso Side Guide (L)
5 Torso Side Guide (R)
6 Back Panel (Front)
7 Pelvis Side Guide (L)
8 Pelvis Side Guide (R)
9 Seat Pan (Anti-Submergence)
10 Seat Pan Base Frame
11 Dual Rail – Left (S1)
12 Dual Rail – Right (S2)
13 Front Hinge Assembly (L)
14 Front Hinge Assembly (R)
15 Rear Hinge / Pinset (L)
16 Rear Hinge / Pinset (R)
17 Energy Absorber – Left
18 Energy Absorber – Right
19 Side Crush Element (L)
20 Side Crush Element (R)
21 Locking Module (L)
22 Locking Module (R)

2. ENERGY ABSORPTION MODULE – EXPLODED

1 Cartridge Housing
2 Guide Rod
3 Piston
4 Main Coil Spring
5 Hydraulic Damper
6 End Cap
7 Preload Adjuster
8 Mounting Clevis

420 (Extended)
240 (Compressed)

FORCE-STROKE CHARACTERISTIC (TYP.)

Stroke (mm)	Force (kN)
0	0
50	10
100	15
150	18
200	20

ENERGY ABSORBER PERFORMANCE (Per Side)

Peak Force @ 25.4 g	Stroke	Energy Absorption Capacity	Operating Temp.
18 – 22 kN	180 mm	2.0 – 2.8 kJ	-40°C to +80°C

3. SIDE CRUSH ELEMENT – EXPLODED

1 19/20A End Plate (Top)
2 19/20B Honeycomb Core
3 19/20C End Plate (Bottom)
4 19/20D Side Bracket
5 19/20E Mounting Bolt

CRUSH ELEMENT PERFORMANCE

Peak Force	Stroke	Energy Absorption	Density (Core)
12 – 15 kN	60 – 80 mm	0.6 – 1.0 kJ	> 120 kg/m³

4. LOCKING MODULE – EXPLODED & STATES

1 2V/22A Housing/Vale
2 2V/22B Tension Spring
3 2V/22C Catch Weight
4 2V/22D Release Lever
5 2V/22E Status Indicator

LOCK STATES & FUNCTION

STATE 0 DRIVE (UNLOCKED)	STATE 1 PREPARE (ARMED)	STATE 2 RIDE DOWN (LOCK POSITION)	STATE 3 END-STOP (SECURE)
Free motion All DOF active Comfort mode	Inertial weight Armed Preloaders	Pawl engaged Guides locked Load path active	End-stop engaged Full lock Crash load secured

INERTIAL TRIGGER SPEC

Trigger Accel.	Engage Time	Reset	Safety Factor
> 2.5 g	< 20 ms	Manual	> 2.0

5. PITCH MECHANISM – EXPLODED

1 Pitch Stop Adjuster
2 Pitch Side Plate L/R
3 Pitch Shaft
4 Section Gear
5 Lock Pin

PITCH POSITIONS

Position	Angle
1	-10°
2	0°
3	+25°

6. RAIL SYSTEM – DUAL RAIL (S1 / S2)

530
480
160 (Stroke)

1 19/22A Outer Rail
2 19/22B Inner Rail
3 19/22C Rail Cage
4 19/22D End Stop
5 19/22E Lubrication Port Grease Filling

7. CONNECTION DETAILS

DETAIL C: FRONT HINGE
DETAIL D: DUAL RAIL MOUNT
DETAIL E: ABSORBER MOUNT

M12 BOLT – CLASS 10.9
TORQUE: 90 Nm
BUSHING ID: M12 x 2.0

M16 BOLT – CLASS 8.8
TORQUE: 85 Nm
WELD: 100% RT-100

M12 BOLT – CLASS 10.9
TORQUE: 95 Nm
BUSHING ID: M12 x 2.0

8. WELDING SPECIFICATION

JOINT TYPE	WELD TYPE	SIZE	LENGTH	NOTES
Frame to Rail	MIG	6 mm	Continuous	Both Sides
Side Bracket	MIG	5 mm	90 mm	Single S1
Cross Member	MIG	6 mm	Continuous	Both Sides
Seat Pan	MIG	4 mm	Continuous	Single OR
Mounting Tab	MIG	5 mm	40 mm	Single S1

WELD CONSUMABLE: ER70S-6
SURFACE PREP: Grind & Clean before welding

9. TOLERANCE TABLE (UNLESS SPECIFIED)

Linear Dimension (0-300)	±0.5 mm
Linear Dimension (300-1000)	±1.0 mm
Hole Diameter	+0.1
Shaft Diameter	+0.05
Angle	±0.5°
Parallelism	0.5 mm / 300 mm
Perpendicularity	0.5 mm / 300 mm

10. CRASH LOAD PATH SUMMARY

1 Head Support → Upper Frame → Side Guides → Seat Pan
2 Seat Back → Side Links → Side Absorbers → Rails
3 Pelvis → Seat Pan → Rails → Energy Absorbers → Vehicle
4 Seat Pan → Crush Elements → Side Brackets → Rails
5 Restraint Loads → Pelvis/Torso Guides → Frame
6 Rails → Vehicle Floor Anchors (4 Points)

11. VEHICLE ATTACHMENT POINTS (REFERENCE)

400
300
390

4X M16/M12 Anchor
Min Grade 8.8
Recommended Torque: 85-95 Nm

12. NOTES

- This drawing package is a research prototype.
- All dimensions are nominal.
- Do not use for production without validation.
- Validate with FE, sled test, and full vehicle test.
- Comply with FMVSS 202a, 207, 301, and other applicable regulations.
- All components must be inspected and tested per validation plan.

13. REVISION BLOCK

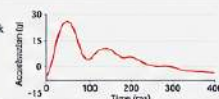
REV.	DATE	DESCRIPTION	BY
A	2025-05-10	Initial Concept Release	ENG
B	2025-05-20	Added Locking Module Details	ENG
C	2025-05-28	Added Sections & Tolerances	ENG
D	2025-06-01	Final Research Release	ENG

T-OCS V5.0 CRASH LOAD PATH ANALYSIS (0-400 ms)

TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT – REAR IMPACT

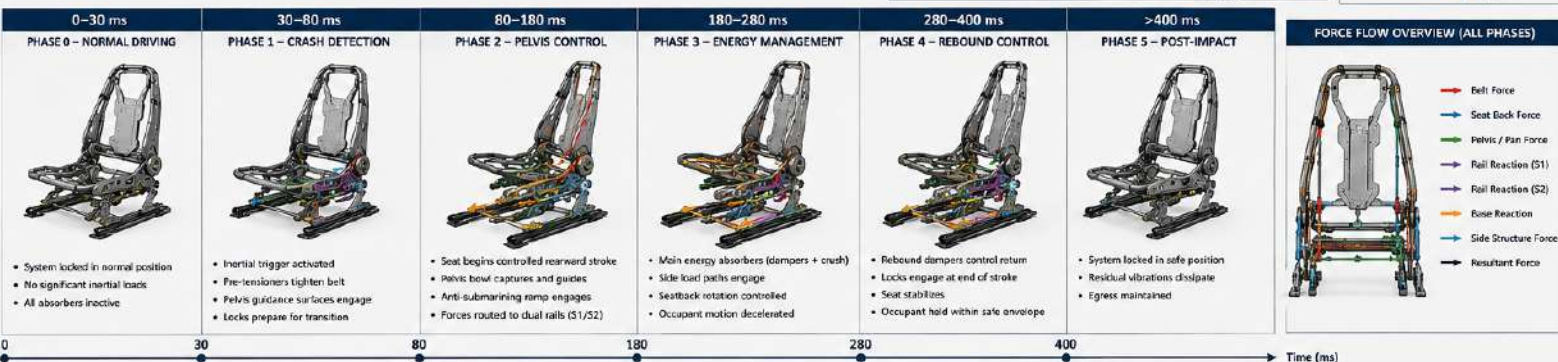
CRASH PULSE (Example)

Scaled FMVSS 301R Pulse – 25.6 g peak
 $\Delta V \approx 61.8$ kph (38.3 mph)
Seat Stroke: 180 mm (Total)

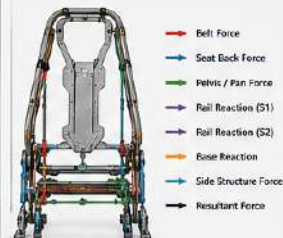


OCCUPANT MODEL

THOR-50M
Belted
50th Percentile Male



FORCE FLOW OVERVIEW (ALL PHASES)



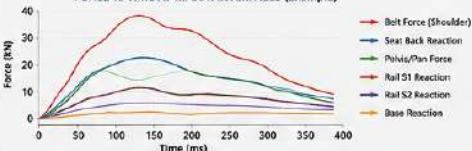
LOAD PATH DETAILS BY SUBSYSTEM

- BELT SYSTEM (Red)**
 - Transfers occupant inertia to torso
 - Shoulder anchorages → Seat structure
- SEAT BACK (Blue)**
 - Occupant back load → Seatback frame
 - Side guides → Rail carriers
- PELVIS / PAN (Green)**
 - Pelvis load → Seat pan
 - Anti-submarining ramp → Side brackets
- RAIL SYSTEM (Purple / Orange)**
 - Loads → S1 & S2 rails
 - Translating carriage → Base
- ENERGY ABSORBERS (Dashed)**
 - Loads → Dampers / Crush elements
 - Convert KE to deformation & heat
- SIDE STRUCTURE (Teal)**
 - Lateral loads → Side links / guides
 - Normalize torsion & yaw
- BASE REACTION (Orange / Brown)**
 - Loads → Base cross member
 - Vehicle floor handprints

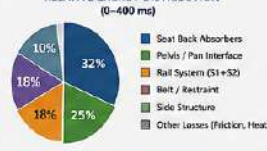
LOAD PATH ANIMATION – KEY TIME INSTANTS



FORCE vs TIME AT MAJOR INTERFACES (Example)



RELATIVE ENERGY DISTRIBUTION (0-400 ms)



CRITICAL LOAD PATHS (DESIGN CHECK)



DESIGN CONSIDERATIONS

- ✓ Redundant load paths for survivability
- ✓ Progressive energy absorption
- ✓ Controlled motion & rotation
- ✓ Minimized occupantamping
- ✓ Load distribution, not load spike
- ✓ Compliant first rigid second
- ✓ Locking at end of stroke
- ✓ Rebound control to prevent secondary impact

PERFORMANCE OBJECTIVES

Max Seat Back Rotation: $\leq 25^\circ$
Seat Stroke (Total): 180 mm
Peak Belt Force: ≤ 15 kN
Pelvis Load (Peak): ≤ 12 kN
Peak Rail Load (Each): ≤ 20 kN
HC (1st Row, 50M): ≤ 500
Hij (neck Tension): ≤ 1.0
Head Excursion: ≤ 700 mm

MATERIALS & PROPERTIES (TYPICAL)

Component	Material	Property	Thickness (mm)
Seat Back Frame	DP680 Steel	Yield 980 MPa	2.0
Seat Pan	DP780 Steel	Yield 780 MPa	1.8
Rails (S1/S2)	Q&P 22MnES	Yield 1200 MPa	2.5
Cross Members	DP550 Steel	Yield 550 MPa	2.0
Energy Elements	High Damping Steel / Polymer	N/A	N/A
Bolts / Pins	SCM440 (10.9)	Tensile 1040 MPa	—
Bushings	Sintered Steel	—	—

KEY BENEFITS

- ✓ Reduces rear-impact injury risk
- ✓ Limits seatback collapse & rotation
- ✓ Controls occupant trajectory
- ✓ Improves belt & restraint efficiency
- ✓ Meets / Supports FMVSS 202a
- ✓ Modular, serviceable architecture

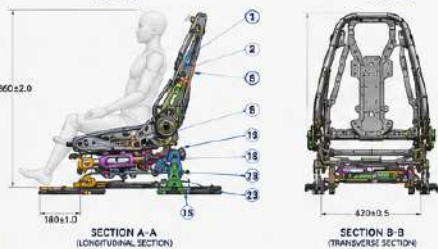
VALIDATION PLAN

- LS-DYNA / ABAQUS FE Simulation
- ATD (THOR, BioRID, Hybrid III) Simulation
- PHMS Correlation (NHTSA / UMTR Protocols)
- Side Testing (FMVSS 301R Pulse)
- Full Vehicle Testing
- DOE & Sensitivity Analysis
- Optimization (Multi-Objective)

NOTES

- All dimensions in mm unless noted
- Tolerances: ± 0.5 mm (unless noted)
- Angles: $\pm 0.5^\circ$
- This drawing is research concept only
- Not for production without validation

500±1.0 520±1.0



ITEM	COMPONENT	MATERIAL	QTY	NOTE
1	Head Support Frame	D9903 Steel	1	
2	Head Side Wing (L)	D9793 Steel	1	
3	Head Side Wing (R)	D9793 Steel	1	
4	Torso Side Guide (Left)	D9793 Steel	1	
5	Torso Side Guide (Right)	D9793 Steel	1	
6	Upper Cross Member	D9903 Steel	1	
7	Back Panel (Compliance)	D9590 Steel	1	
8	Back Recline Hinge (L)	22MnB5 Steel	1	
9	Back Recline Hinge (R)	22MnB5 Steel	1	
10	Pelvis Bolt (Anti-Submerging)	D9590 Steel	1	
11	Pelvis Adjust Actuator (Left)	SCM440	1	Stroke 50 mm
12	Pelvis Adjust Actuator (Right)	SCM440	1	Stroke 50 mm
13	Seat Pan (Adaptive Compliance)	D9590 Steel	1	
14	Pan Front Hinge (Left)	22MnB5 Steel	1	
15	Pan Front Hinge (Right)	22MnB5 Steel	1	
16	Dual Rail System - S1 (Main)	D9903 Steel	2	
17	Dual Rail System - S2 (Redundant)	D9903 Steel	2	
18	Translation Actuator (SUS2)	SCM440	2	Stroke 180 mm
19	Energy Absorber - Damper	6061-T6 Al	2	Hydraulic
20	Energy Absorber - Crush Element	6061 Al	2	Replaceable
21	Base Frame (High Strength Steel)	D9903 Steel	1	
22	Lateral Adjustment Unit (L)	22MnB5 Steel	1	
23	Lateral Adjustment Unit (R)	22MnB5 Steel	1	

RANGE (mm)	TOLERANCE (mm)
0,5 - 3	±0,1
>3 - 6	±0,1
>6 - 30	±0,2
>30 - 120	±0,3
>120 - 400	±0,5
>400 - 1000	±0,8

WELD SPECIFICATIONS
PROCESS: MIG (GMAW)
FILLER: ER70S-6
MIN. FILLET SIZE: 4 mm
WELD CATEGORY: STRUCTURAL
QUALITY LEVEL: ISO 5817 - C

BOLT SPECIFICATIONS
GRADE: 12.9
NUT: LOCKNUT / NYLOC
TIGHTENING TORQUE (Nm)
 M8 - 26 M10 - 55
 M12 - 95 M14 - 150

MATERIAL THICKNESS

DP980:	1.2 – 3.0 mm
DP780:	1.2 – 2.5 mm
DP590:	1.0 – 2.0 mm
22MnB5:	2.0 – 3.0 mm
TUBE DIA:	25 – 32 mm
TUBE THK:	2.0 – 3.0 mm

NOTES:

1. ALL SHARP EDGES TO BE R.T.D MAX.
2. ALL PIPES TO END WITH 2mm RAD. MIN.
3. ALL HOLES TO BE FREE FROM BURRS.
4. ALL WELDS TO BE CONTINUOUS UNLESS OTHERWISE SPECIFIED.
5. SURFACE PROTECTION: E-COAT + POWDER COAT
6. ALL DIMENSIONS CRITICAL TO SAFETY ARE MARKED

T-OCS V5.0
EFFECTIVE GRAVITY SEAT
ENGINEERING DRAWING

06-2005	SCALE: 1:5	DRAWG-V5-G-003-001
---------	------------	--------------------

T-OCS V5.0 – REAR SEAT SYSTEM INTEGRATION
INDUSTRIAL DESIGN – PRODUCTION READY

2+1 REAR SEAT CONFIGURATION



3-SEAT REAR CONFIGURATION



KEY BENEFITS

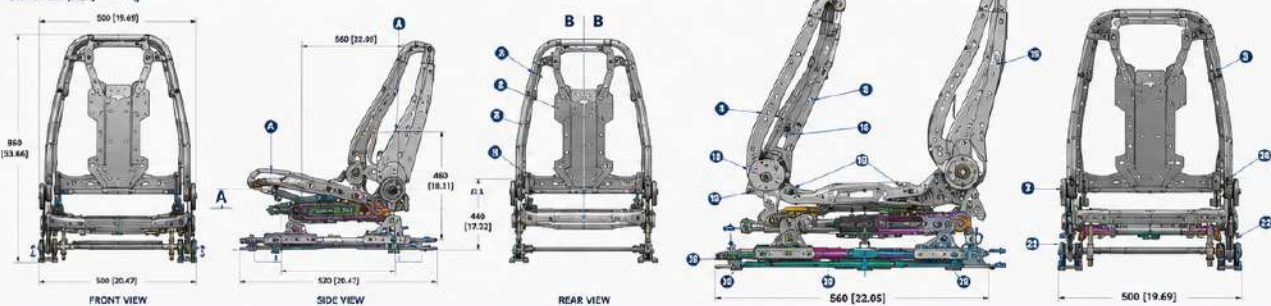
- Reduces occupant ramping and seat rotation
- Manages energy over time (impulse control)
- Lowers head, neck, chest and pelvic injury risk
- Protects rear seat occupants from secondary impact



T-OCS V5.0 ENGINEERING DRAWING – MANUFACTURING PACKAGE

TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT

UNITS: mm [inch]



DETAILS – JOINTS & LOCKING



BILL OF MATERIALS

No.	PART NAME	MATERIAL	QTY
1	Head Support Frame	DP980 Steel	1
2	Head Side Wing (L)	DP780 Steel	1
3	Head Side Wing (R)	DP780 Steel	1
4	Torso Side Guide (L)	DP980 Steel	1
5	Torso Side Guide (R)	DP980 Steel	1
6	Upper Torso Cross Member	DP980 Steel	1
7	Back Panel (Compliance)	DP980 Steel	1
8	Back Recline Hinge (L)	SCM440 Steel	1
9	Back Recline Hinge (R)	SCM440 Steel	1
10	Pelvis Bowl (Anti-Subsiding)	DP780 Steel	1
11	Pelvis Adjust Actuator (L)	Steel + Motor	1
12	Pelvis Adjust Actuator (R)	Steel + Motor	1
13	Seat Pan (Adaptive Compliance)	DP780 Steel	1
14	Pan Front Hinge (L)	SCM440 Steel	1
15	Pan Front Hinge (R)	SCM440 Steel	1
16	Dual Rail System - S1 (Main)	22MnB5 Steel	2
17	Dual Rail System - S2 (Secondary)	22MnB5 Steel	2
18	Translation Actuator (S1/S2)	Steel + Motor	2
19	Energy Absorber - Damper	High Damp Steel	2
20	Energy Absorber - Crush Element	AI 6061-T6	2
21	Base Frame (High Strength Steel)	DP980 Steel	1
22	Lateral Adjustment Unit (L)	Steel	1
23	Lateral Adjustment Unit (R)	Steel	1

GENERAL TOLERANCES (UNLESS SPECIFIED)

- DIMENSIONS : ±0.5
- ANGULAR : ±0.5°
- HOLE DIA : ±0.1
- PROFILE : ±0.5

WELDING

- FLLET WELD SIZE 6 mm (MIN)
- WELD CLASS : ISO 5817 - C
- WELD MATERIAL : ER70S-6

T-OCS V5.0 – REAR SEAT SYSTEM IN VEHICLE (INDUSTRIAL DESIGN – PRODUCTION READY)



SYSTEM SPECIFICATIONS (NOMINAL)	MATERIAL SUMMARY	INDUSTRY STANDARDS / VALIDATION	AVAILABLE FINISHES	APPLICATIONS	NOTES
Total Weight: 29.8 kg ±1.0 kg Ride-Down Stroke (D): 180 mm Vertical Stroke (Z): 60 mm Seat Back Rotation (S): ±25° (Controlled) Seat Pan Roll (P): ±10° Operating Temp.: -40°C to +80°C Dynamic Load Capacity: > 15 kN Life Cycle: 15 Years / 200k Cycles	High Strength Steel: DP980 / DP780 / DP590 Rails: 22MnB5 / Q&P Steel Hinges & Pins: SCM440 (H) Aluminum Parts: 6061-T6 Fasteners: Class 9.8 / 10.9 Dampers: High Damping Steel Crush Elements: Aluminum Honeycomb	✓ FMVSS 202a Compliance (Head Restraints) ✓ FMVSS 207 Integrity (Seat Back Strength) ✓ FMVSS 301R Pulse (Rear Impact) ✓ LS-DYNA / ABAQUS FE Simulation Protocols ✓ PMHS / ATD Correlation ✓ Sled Testing	Black Leather, Dark Brown, Grey Leather, Beige Leather, Alcantara, Perforated Leather, Carbon Style, Brushed Metal	Sedans, SUVs / Crossovers, MPVs / Minivans, Autonomous Vehicles, Premium / Luxury, Future Mobility Platforms	Research prototype - not certified. Design intent: reduce injury risk in rear impacts by controlling occupant kinematics through time-programmed effective force field. All dimensions in mm [inch]. Do not use for production without validation.

T-OCS V5.0 TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT

Exploded View & Mechanism Details

A. EXPLODED VIEW - DETAILED CAD

- Head Support Frame
- Head Support Actuator (Linear)
- Torso Side Guide (Left)
- Torso Side Guide (Right)
- Upper Torso Cross Member
- Back Panel (Compliance)
- Back Recline Hinge (Left)
- Back Recline Hinge (Right)
- Pelvis Bowl (Anti-Submarining)
- Pelvis Adjust Actuator (Left)
- Pelvis Adjust Actuator (Right)
- Seat Pan (Adaptive Compliance)
- Pan Front Hinge (Left)
- Pan Front Hinge (Right)
- Dual Rail System - S1 (Main)
- Dual Rail System - S2 (Redundant)
- Translation Actuator (S1/S2)
- Energy Absorber - Damper
- Energy Absorber - Crush Element
- Base Frame (High Strength Steel)
- Lateral Adjustment Unit (Left)
- Lateral Adjustment Unit (Right)
- Locking Mechanism (Multi-State)



B. MECHANISM SECTIONS & JOINT DETAILS



E. JOINT DEGREES OF FREEDOM

Axis ID	الترتيب	DOF	الحدود
J1	مفصل دوران الظهر (Pivot)	1	-10° ~ +25°
J2	سلك التوجيه (Roll)	1	-5° ~ +5°
J3	سلك التوجيه (Pitch)	1	0° ~ 160°
J4	الترتيب (Yaw)	1	0° ~ 60°
J5	مفصل التوجيه (Pitch)	1	0° ~ 60°
J6	مفصل التوجيه (Pitch)	1	0° ~ 60°

F. MATERIALS

Part Category	Material	Property
Seat Back Frame	DP980 / Q&P Steel	High Strength
Seat Pan	DP780 / A-HSS	High Strength
Rails	22MnB5 / Q&P Steel	High Strength
Cross Members	DP590	High Strength
Energy Elements	High-Damping Steel / Al	Energy Absorption
Bolts / Pins	SCM440 / 12.9 Grade	High Strength

G. TECHNICAL SPECIFICATIONS (NOMINAL)

Parameter	Value (Nominal)
Rearward Stroke	1201
Vertical Stroke (Zin 24)	80 mm
Seat Back Rotation (R)	-10° ~ +25°
Seat Pan Rot (R)	±5°
System Weight Increase	±7.5 kg
Added Cost Target	±120 USD
Response Time	≤ 30 ms
Peak Damping Force	25-35 kN

F. VALIDATION PLAN

- LS-OPT / ADAMS/FE Simulation
- ATD (THOR, BUND, Hybrid III)
- FMVSS Correlation (NHTSA / JNTR)
- Shed Testing (FMVSS 301R Pulse)
- Full Vehicle Testing
- DOE & Sensitivity Analysis
- Optimization (Multi-Objective)

T-OCS V5.0 - INDUSTRIAL DESIGN | جاهز للتصنيع - تصميم صناعي واقعي

A. REAR SEAT - 2+1 CONFIGURATION



B. REAR SEAT - 3-SEAT CONFIGURATION



C. DESIGN FEATURES



D. DIMENSIONS & PACKAGE



E. COLORS & FINISHES



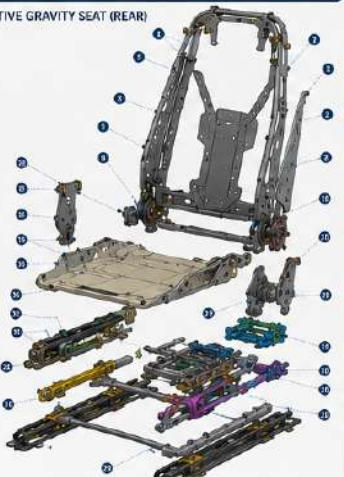
F. APPLICATIONS



T-OCS V5.0 – DETAILED CAD ENGINEERING DRAWING
EXPLODED VIEW – COMPLETE ASSEMBLY

TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT (REAR)

NO.	COMPONENT	QTY
1	Head Support Frame	1
2	Head Side Wing (L)	1
3	Head Side Wing (R)	1
4	Head Support Actuator	2
5	Torso Side Guide (L)	1
6	Torso Side Guide (R)	1
7	Upper Torso Cross Member	1
8	Back Panel (Compliance)	1
9	Back-Reccline Hinge (L)	1
10	Back-Reccline Hinge (R)	1
11	Pelvis Bowl (Anti-Submarining)	1
12	Pelvis Adjust Actuator (L)	1
13	Pelvis Adjust Actuator (R)	1
14	Seat Pan (Adaptive Compliance)	1
15	Pan Front Hinge (L)	1
16	Pan Front Hinge (R)	1
17	Dual Rail System - S1 (Active)	2
18	Dual Rail System - S2 (Rebound)	2
19	Translation Actuator (S1/S2)	2
20	Energy Absorber - Damper	2
21	Energy Absorber - Crush Element	2
22	Base Frame (High Strength Steel)	1
23	Lateral Adjustment Unit (L)	1
24	Lateral Adjustment Unit (R)	1
25	Locking Mechanism (Multi-Stage)	4
26	Centering Mechanism	1
27	Seat Belt Anchor (Integrated)	2
28	Sensor & Control Module	1
29	Wiring Harness Channel	1



MAJOR JOINTS & DETAILS

A BACK RECLINE HINGE



- Dual axis rotation
- Friction control hinge
- Angle range: -10° to 15°

B PAN FRONT HINGE



- Dual link mechanism
- Load path optimized
- Angle range: -5° to 15°

C RAIL LOCKING MECHANISM



- Multi-state lock
- UNLOCK / PREPARE
- LOCK / RIDE-DOWN
- REBOUND LOCK

D TRANSLATION ACTUATOR



- Ball screw + servo motor
- Stroke: 160 mm
- Max Load: 15 kN

E LATERAL ADJUSTMENT UNIT



- Screw drive + guide
- Stroke: <50 mm
- Centering function

SECTION VIEWS



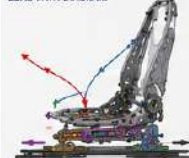
SECTION Y-Y (TRANSVERSE)



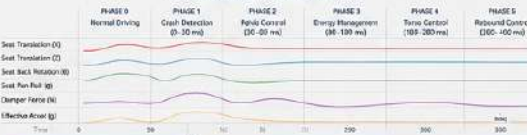
SECTION Z-Z (RAIL SYSTEM)



LOAD PATH DIAGRAM



TIME-PHASED CONTROL STRATEGY



MATERIAL & SPECIFICATIONS

COMPONENT	MATERIAL	PROPERTY
Seat Back Frame	DP600 / Hot Stamp	High Strength
Seat Pan	DP780 / AHSS	High Strength
Rails	22MnB5 / Q&P Steel	High Strength
Cross Members	DP590	High Strength
Energy Elements	High Damping Steel / Aluminum / Polymer	Energy Absorption
Bolts / Pins	SCM440 / 12.9 Grade	High Strength

T-OCS V5.0 – INDUSTRIAL DESIGN (PRODUCTION READY)
TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT – REAR SEAT SYSTEM

T-OCS

2+1 REAR SEAT CONFIGURATION



3-SEAT REAR CONFIGURATION



ADAPTIVE PELVIS CONTROL



TORSO SIDE GUIDES



DUAL RAIL TRANSLATION



ENERGY MANAGEMENT



TECHNICAL SPECIFICATIONS (NOMINAL)

PARAMETER	VALUE (TARGET)
Rebound Stroke (R)	180 mm
Vertical Stroke (Z)	80 mm
Seat Back Rotation (B)	-10° to +15°
Seat Pan Roll (R)	±10°
System Weight Increase	≤ 7.5 kg
Added Cost Target	≤ 130 USD
Response Time	≤ 20 ms
Peak Clamping Force	25-35 kN

PERFORMANCE TARGETS

METRIC	TARGET
Pelvis Submerging	Prevented
Head Excursion (L)	Restraint ≥ 30%
HEIC	Reduced ≤ 30%
Neck NG	Restraint ≥ 30%
Chest Deflection	Reduced ≤ 25%
Seat Back Rotation	Controlled ≤ 35°
FMVSS 202a Compliance	Pass

VALIDATION PLAN

- LS-DYNA / ABACUS FE Simulation
- ATD (THOR, BioRID, Hybrid III) Simulation
- PNA-G Correlation (NHTSA / UMTRI Protocols)
- Sled Testing (FMVSS 201E Pulse)
- Full Vehicle Testing
- ECE & Sensitivity Analysis
- Optimization (Auto-Objective)

KEY BENEFITS

- Active occupant kinematics control
- Reduced injury risk in rear impacts
- Adaptive comfort in normal driving
- Meets FMVSS 202a & global standards
- Modular & scalable architecture

COLOR & MATERIAL OPTIONS



DIMENSIONS (mm)



APPLICATIONS

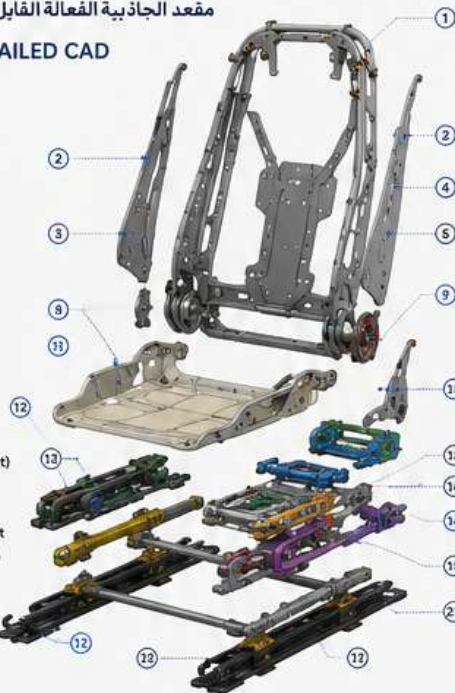
- Sedans
- SUVs
- Crossovers
- MPVs
- Future ADAS / Autonomous Ready
- Compatible with Advanced Restraints

T-OCS V5.0 TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT

مقعد الجاذبية الفعالة القابل للبرمجة زمنياً (الإصدار 5.0)

EXPLODED VIEW – DETAILED CAD

- 1 Head Support Frame
- 2 Head Support Actuator
- 3 Torso Side Guide (Left)
- 4 Torso Side Guide (Right)
- 5 Upper Torso Cross Member
- 6 Back Panel (Compliance)
- 7 Back Recline Hinge (Left)
- 8 Back Recline Hinge (Right)
- 9 Pelvis Bowl (Anti-Submarining)
- 10 Pelvis Adjust Actuator (Left)
- 11 Pelvis Adjust Actuator (Right)
- 12 Seat Pan (Adaptive Compliance)
- 13 Pan Front Hinge (Left)
- 14 Pan Front Hinge (Right)
- 15 Dual Rail System – S1 (Main)
- 16 Dual Rail System – S2 (Redundant)
- 17 Translation Actuator (S1/S2)
- 18 Energy Absorber – Damper
- 19 Energy Absorber – Crush Element
- 20 Base Frame (High Strength Steel)
- 21 Lateral Adjustment Unit (Left)
- 22 Lateral Adjustment Unit (Right)
- 23 Locking Mechanism (Multi-State)



الإضافات والتعديلات الجديدة

- 1 نظام تحكم الحوض النشط
حوض متكيف يمنع الانزلاق تحت الحزام ويوجه الحوض
- 2 مسارات مزدوجة S1/S2
مساران متوازيان لمزيد من الأمان واستمرارية الحركة
- 3 وحدات امتصاص طاقة متدرجة
دمبرات + عناصر سحق قابلة للبرمجة لامتصاص الطاقة
- 4 القفل متعدد الحالات
قفل ميكانيكي + كهربائي يتحكم في حيزية الحركة
- 5 التحكم الجانبي النشط
توجيه جانبي للذراع والحوض لتقليل الانحراف
- 6 تعديل ميلان الظهر النشط
تغيير زاوية الظهر لحظة بلحظة للتحكم بالذراع بالجذب
- 7 دعم الرأس النشط
دعم جانبي + ارتفاع متكيف + شد مسبق للحزام

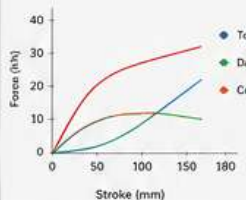
تفاصيل المفاصل والروابط (Joints)



المواصفات الرئيسية

- 7-DOF: عدد درجات الحرية
- 180 mm: مدى الانزلاق الأمامي
- 80 mm: مدى الانزلاق الرأسي
- -10°/+25°: مدى ميلان الظهر
- الوزن: 95-130 kg: قوة قدرتين: المو التصميمية
- المقواد: DP980 – AHSS – 6061 T6
- معايير: FMVSS 202a – ECE R17
- جهد التشغيل: 12-48 V DC

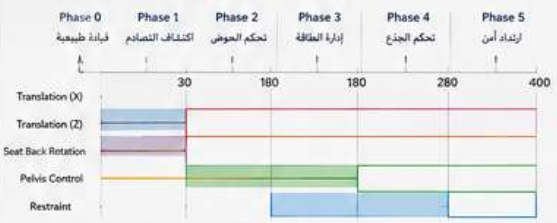
منظومة امتصاص الطاقة



حالات القفل (Lock)

الحالة	الوصف
0	حريتان (قيداً طبيعية)
1	(Propane)
2	قفل الحوض (Lumbar)
3	قفل الانزلاق (Ride-Down)
4	قفل الذراع (Torso Lock)
5	قفل الارتداد (Rebound Lock)

استراتيجية التحكم الزمني



INDUSTRIAL DESIGN – PRODUCTION READY

تصميم صناعي واقعي – جاهز للتصنيع



REAR SEAT CONFIGURATIONS

اتساع الخلفية



المواد والتشطيبات

- إطار رئيسي: DP980 / Q&P Steel
- على السطح: 22MnBS
- عناصر دقيقة: 6061 T6 Aluminum
- عناصر امتصاص: Steel + Polymer
- وسائد: PU Foam + Leather / Fabric
- تشطيب: E-Coat + Powder Coating

مميزات التصميم الصناعي

- تصميم مدمج وخفيف
- تركيب وصيانة سهلة
- تشطيبات عالية الجودة
- تكامل مع أنظمة السيارة
- راحة عالية ونهوية محسنة
- توافق مع جميع فئات المركبات

الأبعاد الرئيسية (مم)



خيارات الألوان والخامات



حلول المشكلات الرئيسية

المشكلة	T-OCS V5.0 - حلول
طيران الجسم للأمام	التحكم الزمني في الانزلاق + تثبيت الحوض + إدارة الطاقة
الانزلاق تحت الحزام الحزام (Submarining)	حوض مقعر + وسائد جانبية + شد مسبق للحزام
إصابة الصدر والظهر	توجيه الذراع الجانبي + تغيير زاوية الظهر + أحزمة متكيفة
إصابة الرأس	دعم رأس نشط + وسادة هوائية جانبية للرأس (اختياري)
الذروة العالية للقوى	عناصر امتصاص طاقة متدرجة + دمبرات قابلة للبرمجة
عدم توافق أحجام الأجسام	تعديل تلقائي للحوض والظهر + حساسات وزن وموضع
الاعتمادية والسلامة	مساران مزدوجان + قفل متعدد الحالات + حساسات مراقبة

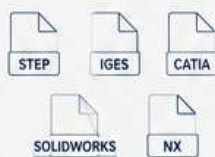
مخطط مسار القوى الفعالة



- قوة الحزام
- قوة الظهر
- قوة الحوض
- رد فعل السكة
- رد فعل القاعدة
- القوة المحصلة

مخرجات CAD – جاهز للتصنيع

النماذج ثلاثية الأبعاد



للوصول إلى نموذج تفاعلي ثلاثي الأبعاد

الرسومات الهندسية

- 2D DRAWINGS (GD&T)
- ASSEMBLY DRAWING
- BOM – PART LIST



توفر أيضاً

- نموذج FEA
- تحليل التصادم
- تقارير الاختبار الافتراضي
- ملفات التصنيع CNC



سلامة أعلى



راحة أعلى



تحكم ذكي



تصميم قابل للتصنيع



قابل للتكامل

T-OCS V5.0

نظام مقعد ذكي للأمان والتحكم بالحركة

T-OCS V5.0 TIME-PROGRAMMABLE OCCUPANT CONTROL SEAT

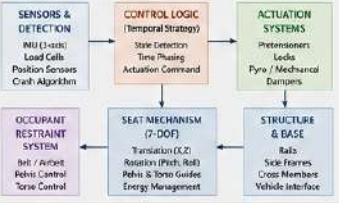
MULTI-DOF DYNAMIC SEAT WITH EFFECTIVE FORCE FIELD CONTROL FOR REAR IMPACT PROTECTION

1. CONCEPT OVERVIEW

T-OCS V5.0 actively manages occupant kinematics by generating time-controlled effective forces through coordinated translation, rotation, adaptive compliance, and restraint interaction to reduce injury risk in rear impact events.



2. SYSTEM ARCHITECTURE



3. KEY MECHANISMS

- DUAL RAIL TRANSLATION UNIT**
S1: Main Slide Drive Rail (180 mm stroke)
S2: Redundant Rail for Load Sharing
- PITCH ROTATION MODULE**
Seat back rotation controlled by dual-side rotary dampers
- PELVIS CONTROL MODULE**
Anti-submerging pan with adaptive pelvic bowl and belt load path guides
- TORSO CONTROL MODULE**
Side thorax guides with compliance adjustment surfaces
- ENERGY MANAGEMENT MODULE**
Progressive dampers + crush elements for controlled energy absorption
- HEAD SUPPORT SYSTEM**
Active adjustable headrest with side wings and pre-tensioning
- LATERAL ADJUSTMENT MODULE**
Occupant centering and lateral position control
- LOCKING & SAFETY MODULE**
Multi-state locking for each phase with fail-safe design

9. TECHNICAL SPECIFICATIONS (NOMINAL)

PARAMETER	VALUE (TARGET)
Reinforced Stroke (X)	180 mm
Vertical Stroke (Z)	80 mm
Seat Back Rotation (θ)	0° to 25° controlled
Seat Pan Roll (φ)	±10°
System Weight Increase	±7.5 kg
Added Cost Target	≤ \$120 USD
Response Time	≤ 30 ms
Peak Damping Force	25-35 kN

10. PERFORMANCE TARGETS

METRIC	TARGET
Pelvis Submerging	Prevented
Head Excursion (X)	Reduced ≥ 30%
HIC15	Reduced ≥ 30%
Neck Mj	Reduced ≥ 30%
Chest Deflection	Reduced ≥ 25%
Seat Back Rotation	Controlled ± 25°
FMVSS 202a Compliance	Pass

11. VALIDATION PLAN

- LS DYNA / ABAQUS FE Simulation
- ATD (THOR, BioRID, Hybrid III) Simulation
- PMHS Correlation (NHTSA / UNTTRI Protocols)
- Sled Testing (FMVSS 301R Pulse)
- Full Vehicle Testing
- DOE & Sensitivity Analysis
- Optimization (Multi-Objective)

12. OCCUPANT MODELS

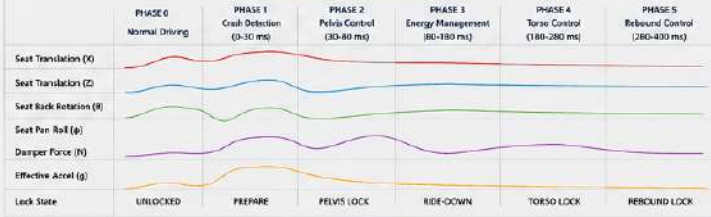


13. APPLICABILITY

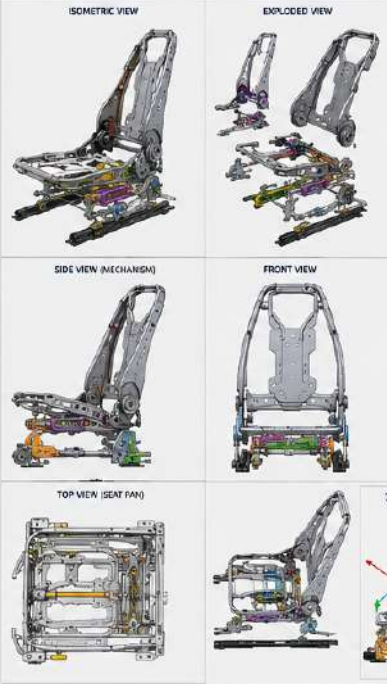
- Front Row – Driver & Passenger
- Rear Row – All Occupants
- Sedans, SUVs, Trucks
- Future ADAS / Autonomous Ready
- Compatible with Advanced Restraints



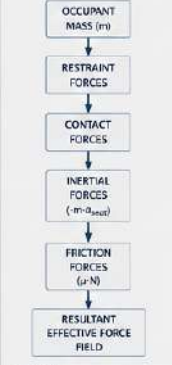
4. TIME-PHASED CONTROL STRATEGY



5. MULTI-DOF MECHANISM DETAILS



6. EFFECTIVE FORCE SYSTEM



$$F_{eff} = m(g - a_{seat}) + F_{contact} + F_{restraint} + F_{friction}$$

7. LOAD PATH DIAGRAM



8. MATERIAL STRATEGY

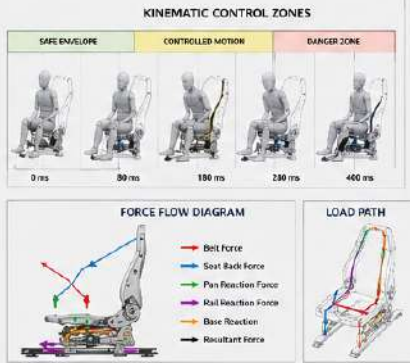
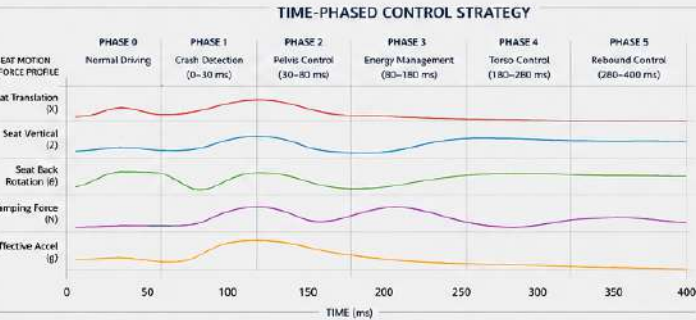
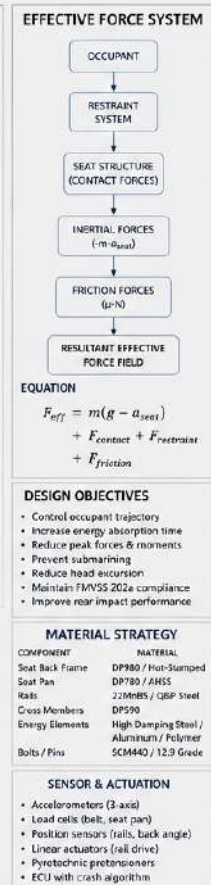
COMPONENT	MATERIAL	PROPERTY
Seat Back Frame	DP980 / Hot Stamped	High Strength
Seat Pan	DP780 / AHSS	High Strength
Rails	22MnB5 / Q&P Steel	High Strength
Cross Members	DP590	High Strength
Energy Elements	High Damping Steel / Aluminum / Polymer	Energy Absorption
Bolts / Pins	SCM440 / 12.9 Grade	High Strength

TIME-PROGRAMMABLE EFFECTIVE GRAVITY SEAT (T-EGS)

Dynamic Force-Field Seat for Rear Impact Protection

CONCEPT

A multi-DOF smart seat that actively manages occupant kinematics by generating time-controlled effective forces through coordinated translation, rotation, adaptive compliance, and restraint interaction to reduce injury risk in rear impacts.



TECHNICAL SPECIFICATIONS	
PARAMETER	VALUE (TARGET)
Rearward Stroke (X)	120 mm
Vertical Stroke (Z)	60 mm
Seat Back Rotation (R)	±25°
Seat Pan Roll (φ)	±10°
System Weight Increase	≤ 6.5 kg
Added Cost Target	≤ 110 USD
Response Time	≤ 30 ms
Peak Damping Force	25-35 kN

PERFORMANCE TARGETS	
METRIC	TARGET
Pelvis Submarining	Prevented
Head Excursion (Δ)	Reduced ≥30%
HCIS	Reduced ≥30%
Neck Nij	Reduced ≥30%
Chest Deflection	Reduced ≥25%
Seat Back Rotation	Controlled ≤ 10°
FMVSS 202a Compliance	Pass

VALIDATION PLAN	
• LS-DYNA / ABAQUS FE Simulation	
• ATD (THOR, BioRID, Hybrid III) Simulation	
• PNAIS Correlation (NI-TSA / UMTR Protocols)	
• Sled Testing (FMVSS 301R Pulse)	
• Full Vehicle Testing	
• DOE & Sensitivity Analysis	
• Optimization (Multi-Objective)	



APPLICABILITY	
• Front Row – Driver & Passenger	
• Rear Row – All Occupants	
• Sedans, SUVs, Trucks	
• Future ADAS / Autonomous Ready	
• Compatible with Advanced Restraints	